01 DocShield AI

AI Document Authenticity Screening

1. Problem and users

Official documents are falsified through edited text, replaced photos, forged stamps, altered QR codes — and now through AI generation. Manual review is slow, repetitive, and misses precisely the subtle cases.

Who. Teams that verify documents at volume: recruitment, finance, compliance and administration.

Positioning

An AI-assisted screening tool that returns **evidence a reviewer can act on** — not a verdict. Every flag is drawn and numbered on the document itself, so the reviewer sees *where* to look, not only *that* something is wrong.

2. Evidence

Tests run against the working prototype

1. AI-generated invoice, visually flawless → scored *low risk*.
2. Client tax identifier of invalid length → missed by the model.
3. Low score returned together with a reject verdict → contradictory output.

Each finding changed the build. An external tester at TecForge then ran three documents through the prototype and reported four further defects — all corrected within the challenge window. See deliverable 03.

3. Solution — a division of labour

Upload → Normalization → Multimodal analysis → Deterministic checks → Risk score → Report

Handled by the model

- ✓ **Reads** the document and extracts fields.
- ✓ **Sees** visual manipulation: mixed fonts, broken alignment, re-inserted text.
- ✓ **Explains** each finding in plain language.

Never trusted to the model

- × **Counting digits** — identifier length.
- × **Checksums** — IBAN mod-97 key.
- × **Arithmetic** — VAT and column totals.

Handled by deterministic code: exact and auditable.

4. Prototype — live

- PDF, JPG, PNG. Every page is rendered and sent, so an inconsistency between two pages of one document is visible to the model. Preview and model input share the same images, so a flagged region always lands where the reviewer looks.
- Strict JSON schema, re-validated server-side — a malformed analysis is rejected, never half-displayed.
- A failed deterministic check raises the score and blocks an "accept" verdict.

Out of 72h scope, part of the target architecture: OCR tuning, dedicated manipulation models, metadata forensics, cross-document checks.

5. Validation and business logic

Focus: how documents are verified today, which frauds are hardest to catch, and what evidence earns a reviewer's trust. A learning log records who was contacted, what was learned, what was surprising, and what changed.

Customer Organizations verifying documents at volume — lenders, insurers, recruiters, schools, administrations.
Value Less manual screening, faster triage, traceable evidence.
Model B2B SaaS per seat or per document, plus an integration API.
Market Finance, recruitment, insurance, education, public sector.
Next 30 days Pilot with one organization, add metadata forensics and cross-document consistency, test pricing.

Key assumption

Organizations will adopt AI-assisted screening only if it shows transparent evidence for every flag. A score without justification will not be trusted — and should not be.